Strategy-proof Market Segmentation against Price Discrimination*

ZHONGHONG KUANG — *Renmin U*
SANXI LI — *Renmin U*
YI LIU — *Yale*
YANG YU — *CUP*

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Abstract

In light of prevailing data regulations, consumer mobility across diverse markets inherently endogenizes market segmentation. Considering such strategic interactions, we define a market segmentation as strategy-proof when no consumer (with positive measure) has an incentive to deviate to another market. We show that in every strategy-proof market segmentation, the producer surplus remains at the uniform monopoly level, and the consumer surplus is bounded between the buyer-optimal level and the uniform monopoly level. Remarkably, no consumer is worse off than in the case of a uniform monopoly. We also construct a family of strategy-proof segmentations to realize every possible welfare outcome.

Keywords: Price discrimination, Market segmentation, Strategic consumer, Welfare;

JEL Classification: D42, D83, L12

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1 Introduction

Online sellers can leverage consumer data for market segmentation and price discrimination, potentially harming consumer welfare. Consequently, scholars and competition authorities have proposed the prohibition of price discrimination as a straightforward means to safeguard consumers.

Recent privacy protection advancements like GDPR offer an alternative approach: enabling consumers to strategically switch among market segments, which potentially enhances consumer surplus. The first step of such mobility, *the Right to be Forgotten*, is already legally mandated.[1] Considering consumers' decentralized, self-interested, and insignificant nature, can they be effectively protected against a monopolist who engages in price discrimination? To explore this issue, we study an extreme scenario where consumers face no costs when changing the market segment. This would provide a (non-cooperative) decentralized foundation for the market segmentation.

In our model, a monopolistic producer sells a homogeneous product to a large number of non-negligible consumers, each with a unit demand. The production exhibits constant marginal cost, which can be normalized to zero. The reservation price of each consumer takes the value from a finite set. The timing is as follows. In the first stage, each consumer simultaneously selects one market to join. In the second stage, the producer sees the exact composition of each market segment and offers different take-it-or-leave-it prices in each market. Finally, each consumer makes the purchase decision.

To reconcile the "large number" and "non-negligible" characteristics of consumers, we model the consumer population as a continuum. Meanwhile, we say a *profitable unilateral deviation* exists if a consumer with an infinitesimal positive measure can shift to a different market segment to achieve a strictly higher payoff. Here, the unilateral deviation can change the prices in markets since sellers online typically adopt algorithmic pricing (see e.g., Johnson et al. (2023)), and consumers are rational in realizing this fact. We define a market segmentation as *strategy-proof* if it satisfies two conditions: (i) the producer's strategy assigns an optimal price in every market segment (including off-path ones), and (ii) no profitable unilateral deviations are possible for consumers.

The seller can only implement third-degree price discrimination. Another interpretation is that the seller possesses precise knowledge of the consumers' valuations but faces regulatory constraints that prevent perfect price discrimination. Instead, the seller is restricted to setting uniform prices within each market segment. For example, certain legal provisions in jurisdictions such as New York and

[1] All iOS apps on the App Store are required to offer an in-app account deletion option by July 30, 2022 (see https://developer.apple.com/support/offering-account-deletion-in-your-app). Moreover, all iOS apps should let people use them without a login (see App Store Review Guideline 5.1.1(v) in https://developer.apple.com/app-store/review/guidelines/#data-collection-and-storage).

California prohibit the use of specific personal characteristics (e.g., gender) for pricing purposes, even when these traits are correlated with willingness to pay or when individual valuations are observable. Moreover, product prices on Steam are tailored based on users' IP addresses with no intra-regional price discrimination. In this case, consumers can alter their IP addresses (to low-income countries) through virtual private networks (VPNs) to receive a lower price.[2]

Our main result is the characterization of the welfare consequences of all strategy-proof market segmentations. With consumers engaging in strategically switching market segments before the producer sets the price, producer surplus is always fixed at the uniform monopoly profit, while consumer surplus can range from the uniform monopoly surplus to the maximum consumer surplus attainable under arbitrary segmentation (known as the *buyer-optimal* outcome). Moreover, no consumer would face a higher price than the uniform monopoly price, suggesting that all consumers are weakly better off.

We know that at least the uniform monopoly and buyer-optimal outcomes are achievable under strategy-proof market segmentation. When there is no market segmentation, the producer will charge the uniform monopoly price. Any deviation would require creating a new market, which is not profitable for a consumer. For the buyer-optimal outcome, we can partition the aggregate market into several *extremal markets*, where every valuation on the support is optimal. The producer merely charges the minimum valuation in each market.[3] Prices in extremal markets are sensitive to entry, which enables strategy-proof segmentation to have different prices for different markets.

With the above two attainable outcomes, we construct a family of strategy-proof segmentation to implement any outcome in between.[4] We start with the buyer-optimal segmentation mentioned above. We then merge markets with the highest prices to create a new market with a share equal to a parameter between zero and one. Assuming that the critical market can be proportionally divided, the consumer surplus in the construction is continuous in the parameter. Moreover, two extreme cases of this construction correspond to the uniform monopoly and buyer-optimal outcomes, respectively. Thus, any consumer surplus in between can be obtained.

Our work springs from two strands of literature. First, our paper primarily belongs to the literature studying the welfare consequences of price discrimination (e.g., Aguirre et al. (2010) and Cowan (2016)). In a seminal paper, Bergemann et al. (2015) characterizes all possible welfare consequences in third-price

[2]Additionally, technical constraints may also prevent perfect price discrimination at the individual level. In the Steam example, producers can only aggregate data by IP address, and identifying individual consumers is technically impossible.

[3]Consider all markets whose optimal price equals the minimum possible willingness-to-pay. Since deviation cannot lead to a lower price, all consumers in these markets have no incentive to leave. Therefore, by iteratively considering markets from the lowest price to the highest one, we confirm that the above construction is strategy-proof.

[4]We cannot immediately conclude that any outcome in between is feasible since a convex combination of strategy-proof market segmentation may not necessarily be strategy-proof.

discrimination with exogenous segmentation. The shaded triangle in Figure 1 depicts all available surplus pairs, in which A marks the uniform monopoly outcome and C marks the buyer-optimal segmentation.[5] The line segment between A and C represents the set of possible welfare pairs in our results. Ichihashi and Smolin (2025) investigate how the surplus triangle shrinks when the producer has private information unknown to the central designer.

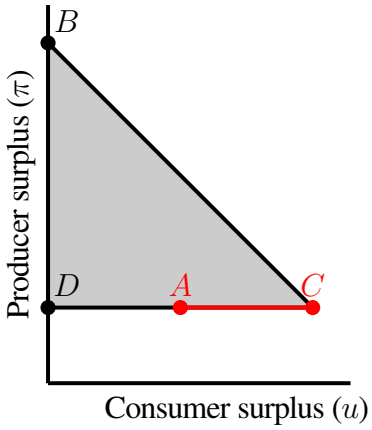

Figure 1: Surplus Triangle

The market segmentation problem analyzed in Bergemann et al. (2015) can also be viewed as an information design problem with a single consumer. Then, Roesler and Szentes (2017) study how an uninformed buyer, with a known prior value distribution, learns the distribution. Condorelli and Szentes (2020) further endogenize the selection of value distribution. These studies mostly focus on settings with a single buyer and a homogeneous product, which is not suitable for endogenizing the market segmentation. Hence, we incorporate a large number of strategic consumers, which complements previous studies.[6]

Second, this paper is also linked to the literature on strategic consumers. In a cooperative setting, Haghpanah and Siegel (2023b) study the formation of market segmentation by consumers, whereas our paper considers the formation as the equilibrium of a non-cooperative game. As a result, the market they analyzed must be efficient, the producer surplus may arise, and the outcome may not be Pareto-improving compared with the aggregate market. In contrast, the market segmentations analyzed in our paper may be inefficient, the producer surplus is fixed, and the outcome is always Pareto-improving.

One mainstream of literature assumes repeated purchases from a monopolist. In such *behavior-based price discrimination* scenarios, the monopolist obtains information about consumers through their

[5]Point B marks the first-degree price discrimination outcome, in which consumers with the same valuations are grouped. Point D marks the outcome where social welfare is minimized.

[6]See Ichihashi (2020); Deb and Roesler (2024); Haghpanah and Siegel (2023a, 2022) for examples of bilateral trade models with multiple products and Armstrong and Zhou (2022); Belleflamme et al. (2020); Elliott et al. (2021); Chen et al. (2020); Rhodes and Zhou (2024) for examples with multiple strategic and competitive producers.

purchase history, and consumers act myopically or strategically against the monopolist (Fudenberg and Tirole, 2000; Gehrig and Stenbacka, 2007; Shen and Villas-Boas, 2018; Bonatti and Cisternas, 2020).[7] Acquisti and Varian (2005) argues that purchase history may harm consumers in later periods, which, in turn, forces consumers to protect their privacy. Chen and Zhang (2009) consider strategic consumers seeking the lowest price when facing dynamic pricing. Our paper differs structurally from those studies, as strategic consumers in our model make only a single purchase decision, while their strategic feature arises from their interaction with other consumers.

The strategic concerns of consumers are also present in their voluntary disclosure decisions before the purchase. For example, Ali et al. (2023) investigate the incentive to voluntarily disclose hard evidence (exact information or a range about personal preferences), which builds the market segmentation. Hidir and Vellodi (2021) study the soft information about consumers' valuations and introduce the incentive-compatible market segmentation. In their settings, the strategic side of consumers is manifested by their decisions on their data, which indirectly influences market segmentation. While in ours, each consumer directly chooses the market segment.

The rest of the paper is organized as follows. Section 2 sets up the model. Section 3 presents the main result and the proof. Section 4 considers costly deviation. Finally, Section 5 discusses an alternative definition to endogenize market segmentation.

2 Model

A monopolistic producer sells a product to N consumers, with a constant marginal cost normalized to zero. Since N should be very large in the real world, we can approximate that the seller serves a continuum of consumers for convenience.[8] Consumers are uniformly distributed within $[0,1]$, each with unit demand. Consumers' willingness-to-pay of the product is determined by a function $V:[0,1]\to\mathbb{R}_+$. The range of V is finite and it is $\{v_1, v_2, \cdots, v_K\}$ with $v_{k-1} < v_k$. We use $\theta \in [0,1]$ to denote a representative consumer.

In this setting, a *market* X is a subset of $[0,1]$. We assume that there are n market segments, indexed by $i=1,...,n$, where $n \geq K$ is exogenously given. A *market segmentation* is a collection of markets $\mathcal{P}=\{X_1,...,X_n\}$ such that $X_i \cap X_j = \emptyset$ for all $i \neq j$, and $\bigcup_{i=1}^n X_i = [0,1]$. Each set X_i represents the

[7] See the detailed survey conducted by Fudenberg and Villas-Boas (2006).

[8] We can alternatively study an N-consumer model directly without approximation. As N grows, the results in an N-consumer model will converge to the results in the approximated model with a continuum of consumers, which can be rigorously proven by the epsilon-N method. However, the continuum approximation is easier to understand since it avoids tedious rounding procedures.

group of consumers who choose segment i. We say that a consumer θ is assigned to segment i if $\theta \in X_i$. The producer observes the composition of each segment and can set different prices across segments.

Timeline. The timeline of the extensive-form game is summarized as follows.

1. Each consumer $\theta \in [0,1]$ chooses a segment $i \in \{1,...,n\}$.
2. This induces a market segmentation $\{X_1,...,X_n\}$, where X_i is the set of consumers assigned to segment i. The producer observes the composition of each segment and sets a price for each segment.
3. Each consumer makes a purchase decision.

Producer's Strategies. In stage 2, the producer offers a take-it-or-leave-it price to maximize the profit in each market. We say $p^* \in \mathbb{R}_+$ is an *optimal price* of market X if

$$p^* \in \operatorname*{argmax}_{p \geq 0} \int_X p \mathbb{I}\{V(\theta) \geq p\} d\theta.$$

Multiple optimal prices might exist for a market. Let ϕ denote the deterministic pricing strategy of the producer, which maps every market to a non-negative price. If $\phi(X)$ is an optimal price of market X for any $X \subseteq [0,1]$, the strategy ϕ is called *rational*.

Consumer's Strategies. We only consider deterministic strategies of consumers. In stage 1, each consumer $\theta \in [0,1]$ independently chooses one of n market segments to join. The segment that consumer θ joins is denoted by $\sigma(\theta) \in \{1,2,\cdots,n\}$. Given strategy profile σ, we say $\mathcal{P}$ is the induced market segmentation if $X_i = \{\theta | \sigma(\theta) = i\}$. In stage 3, consumer θ makes a purchase decision if the price offered by the producer $\phi(X_{\sigma(\theta)})$ is no larger than the willingness-to-pay $V(\theta)$.

Payoffs. Let u, π, and w denote the consumer surplus, producer surplus, and total surplus, respectively. Given producer's strategy ϕ and consumers' strategies σ, the producer surplus is $\sum_{i=1}^n \phi(X_i) \int_{X_i} \mathbb{I}\{V(\theta) \geq \phi(X_i)\} d\theta$. Meanwhile, the payoff of consumer θ is denoted by $u_\theta(\phi,\sigma) = \max\{V(\theta) - \phi(X_{\sigma(\theta)}), 0\}$. Hence, the consumer surplus is $\sum_{i=1}^n \int_{X_i} u_\theta(\phi,\sigma) d\theta$; and the social welfare is $\sum_{i=1}^n \int_{X_i} V(\theta) \mathbb{I}\{V(\theta) \geq \phi(X_i)\} d\theta$.

For the aggregate market, we make a generic assumption that the optimal uniform price v^* is unique. We assume $v^* > v_1$ to avoid trivial analyses. Under the uniform price monopoly, we use u^*, π^* to denote the consumer surplus and the producer surplus, respectively. Let $\overline{w}$ denote the maximum social welfare, i.e. $\int_0^1 V(\theta) d\theta$.

Solution Concept. We first introduce how we model the individual deviation. With N consumers, the market share of an individual consumer is $\varepsilon = \frac{1}{N} \to 0^+$, which is small but non-negligible. When approximating discrete consumers with a continuum framework, each individual consumer in the discrete setting corresponds to a positive but infinitesimal measure of consumers in the continuous setting.[9]

We begin by defining the profitable deviation of a positive measure of consumers. Only when the measure of consumers is positive can such deviation fundamentally change the market segmentation, the prices, the consumer surplus, and the producer surplus.

Definition 1. *Given the strategy profile (ϕ,σ), we call a set of consumers $D \subseteq [0,1]$ a **profitable deviation set**, if (i) the measure of D is positive; (ii) there exists another strategy of consumers σ' such that*

- $\sigma'(\theta) \neq \sigma(\theta)$ *if and only if* $\theta \in D$,
- $u_\theta(\phi,\sigma') - c > u_\theta(\phi,\sigma)$ *for all* $\theta \in D$.

The parameter $c \geq 0$ represents the deviation cost, which is zero during the main analysis. Since we adopt the non-cooperative game-theoretical approach, such a deviation should strictly benefit all consumers in the set D, featuring the strong Nash solution.

We then define the profitable unilateral deviation to capture an individual's deviation.

Definition 2. *We say there is a **profitable unilateral deviation** if for any $\varepsilon > 0$ there exists a deviation set with a measure smaller than ε.*

Now, we can formally define our solution concept.

Definition 3. *Given a strategy profile (ϕ,σ), we say it induces a **strategy-proof** market segmentation $\mathcal{P}$ if (i) ϕ is rational; (ii) there is no profitable unilateral deviation.*

3 The Main Result and the Proof

We first present the main result of this paper and then provide the proof.

Theorem 1. *There exists a strategy profile (ϕ,σ) that induces strategy-proof segmentation **if and only if** producer surplus $\pi = \pi^*$ and consumer surplus $u \in [u^*, \overline{w} - \pi^*]$. No positive measure of consumers is worse off compared with the uniform monopoly outcome.*

[9] This framework can also be understood through the following analogy. In discrete consumer models, each individual is the "smallest" set with a positive market share. Correspondingly, in continuous consumer models, this "smallest" set is represented by an infinitesimal yet positive measure of consumers.

Prior to presenting the comprehensive proof, we illustrate our result and intuitions using a market with consumers having either high (v_H) or low (v_L) willingness-to-pay. To avoid trivial analysis, we assume that v_H is the optimal price without price discrimination. Without loss of generality, the aggregate market is divided into two segments, with consumers freely choosing which segment to join. Let market 1 contain masses α and β of high- and low-type consumers, respectively. If $v_L(\alpha+\beta) \geq \alpha v_H$, the producer can charge price v_L in market 1. Since v_H is the uniform monopoly price, market 2 must include a positive measure of high-type consumers.

We examine whether v_L in market 1 can be maintained. If so, how is a high-type consumer in market 2 deterred from switching to market 1? First, we find that a profitable unilateral deviation exists when $v_L(\alpha+\beta) > \alpha v_H$, because a high-type consumer with an infinitesimal measure can join market 1 without affecting its v_L price. However, when $v_L(\alpha+\beta) = \alpha v_H$, any such deviation triggers a price increase to v_H. Consequently, this equality is necessary for a strategy-proof segmentation with price v_L in market 1. Moreover, for a given β, the entire market segmentation is uniquely pinned down.

As a result, the producer surplus equals that of the uniform monopoly case, as v_H remains optimal in market 1. Meanwhile, the consumer surplus depends on the value of β. At $\beta = 0$, the outcome coincides with the uniform monopoly, while at β equal to the total mass of low-type consumers, the buyer-optimal outcome is achieved. By varying β, the full spectrum of intermediate outcomes between these two extremes is attainable.

Now we provide the detailed proof of our theorem.

Proof for Necessity

For market X, let **supp**$\{X\}$ denote the *support set* of the valuation distribution in market X. Formally, this support set comprises those valuations at which the measure of consumers exhibiting the corresponding willingness-to-pay is strictly positive.

In our illustrative example involving valuations v_H and v_L, we demonstrate that if v_L is optimal in a given market, then v_H must also be optimal. This observation provides key insights into maintaining market segmentation with distinct pricing across different markets. Building on this, we generalize the condition as follows.

Indifference Condition. If there exist two markets with different prices, $\phi(X_i) < \phi(X_j)$, then for all $v_k \in \mathbf{supp}\{X_j\} \cap (\phi(X_i), \phi(X_j)]$, v_k is also optimal in market X_i.

In the following, we prove that the **indifference condition** is necessary to induce a strategy-proof

segmentation when the producer's strategy ϕ is rational.

Lemma 1. *The strategy profile (ϕ,σ) induces a strategy-proof segmentation $\mathcal{P}=\{X_1,\cdots,X_n\}$* ***only if*** *(i) ϕ is* ***rational****; and (ii) the* ***indifference condition*** *holds.*

Proof of Lemma 1. Suppose (ϕ,σ) induces a strategy-proof segmentation $\mathcal{P}=\{X_1,\cdots,X_n\}$. If $\phi(X_i)<\phi(X_j)$ for markets X_i and X_j, for any consumer in market X_j whose value v_k lies in $(\phi(X_i),\phi(X_j)]$, his payoff is zero.

We prove this lemma by contradiction. Suppose v_k is not an optimal price of market X_i, we define

$$r=\phi(X_i)\int_{X_i}\mathbb{I}\{V(\theta)\geq\phi(X_i)\}d\theta-v_k\int_{X_i}\mathbb{I}\{V(\theta)\geq v_k\}d\theta>0.$$

Then, for any $\varepsilon>0$, since the set $\{\theta\in X_j:V(\theta)=v_k\}$ has positive measure, we can chooses a subset $D\subseteq\{\theta\in X_j:V(\theta)=v_k\}$ such that $0<\mu(D)<\min\{\varepsilon,\frac{r}{v_k-\phi(X_i)}\}$, where $\mu(D)$ is the measure of set D. Consider the following possible deviation

$$\sigma'(\theta)=\begin{cases}\sigma(\theta), & \theta\notin D,\\ i, & \theta\in D.\end{cases}$$

Consider the market $X_i'=X_i\cup D$ induced by strategy σ'. The difference of the revenue of pricing $\phi(X_i)$ and the revenue of pricing v_k is $r+\phi(X_i)\mu(D)-v_k\mu(D)>0$. Thus, v_k cannot be the optimal price of market X_i'. Since D only contains consumers with valuation v_k, any $v>v_k$ cannot be the optimal price of market X_i'. Therefore, $\phi(X_i')<v_k$.

This implies that deviation is profitable for all consumers in D, resulting in a contradiction. □

Consider a strategy-proof segmentation $\mathcal{P}=\{X_1,\cdots,X_n\}$ induced by strategy profile (ϕ,σ). Lemma 2, the key step for necessity, states that the maximum price across all markets $(\max\{\phi(X_i)\}_{i=1}^n)$ must equal the uniform monopoly price v^*.

Lemma 2. *For any strategy-proof segmentation $\mathcal{P}=\{X_1,\cdots,X_n\}$, $\max\{\phi(X_i)\}_{i=1,\textbf{supp}\{X_i\}\neq\emptyset}^n=v^*$.*

Proof of Lemma 2. Figure 2 graphically illustrates the main logic to prove Lemma 2. The three panels represent three different pricing strategies. In the upper panel, the producer charges $\phi(X_i)$ in market i. Without loss of generality, we assume $\phi(X_1)\leq\phi(X_2)\leq\cdots\leq\phi(X_n)$. The producer surplus, in this case, is denoted by π. In the middle panel, the producer charges a uniform price $\phi(X_n)$ in all markets. Denote the producer surplus in this case by π'. In the lower panel, the producer charges the optimal uniform price v^* in all markets and obtains the uniform monopoly profit π^*.

We first argue that the three pricing strategies yield the same producer surplus. By Lemma 1 and noticing that $\phi(X_n)\in\textbf{supp}\{X_n\}$, $\phi(X_n)$ should be optimal in every market. Thus, the pricing strategies

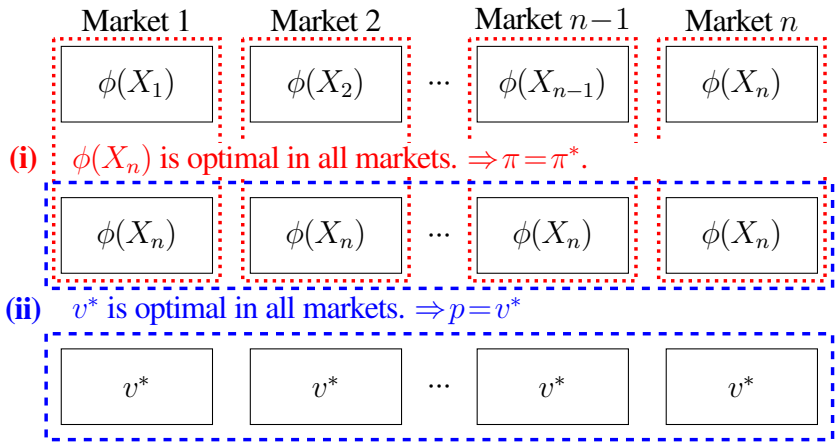

Figure 2: Why $\max\{\phi(X_i)\}_{i=1}^n = v^*$?

in the upper and middle panels result in the same producer surplus, that is, $\pi=\pi'$. Evidently, $\pi'\leq\pi^*$, since the latter is the uniform monopoly profit and the former is the profit by charging a uniform price $\phi(X_n)$. Moreover, from the results of the surplus triangle by Bergemann et al. (2015), we know that $\pi^*\leq\pi$. Thus, we must have $\pi=\pi'=\pi^*$.

Finally, since $\pi'=\pi^*$, $\phi(X_n)$ must be an optimal uniform price, suggesting that $v^*=\phi(X_n)$ by the uniqueness of the optimal uniform price. □

Lemma 2 reveals a fact that circulating consumers prevent the producer from pricing higher than v^*. All results shown in Theorem 1 are direct implications of Lemma 2.

- **Producer surplus**. Since v^* is optimal in all markets within segmentation $\mathcal{P}$, the producer surplus is unchanged if charging v^* to all markets, resulting $\pi=\pi^*$.
- **Pareto improvement**. Zero measure of consumers face a higher price than v^*.
- **Consumer surplus**. By Pareto improvement, consumer surplus is at least the uniform monopoly outcome, $u\geq u^*$. Meanwhile, $u\leq\overline{w}-\pi^*$ by the surplus triangle.

Proof for Sufficiency

We need to construct strategy-proof segmentations to realize every (π^*,u) with $u\in[u^*,\overline{w}-\pi^*]$. For one extreme case, no segmentation can achieve $u=u^*$. The other extreme case ($u=\overline{w}-\pi^*$) is realized by greedy procedures we will explain immediately. During the remaining construction process, we only use the minimum optimal pricing rule $\phi^{\min}$ that maps every market to its minimum optimal price.

To simplify our further construction, we introduce a new perspective on market segmentation. We say $\{\mathbf{x}_1,\cdots,\mathbf{x}_n\}$ where $\mathbf{x}_i\in\mathbb{R}^K$ is a *canonical representation* of market segmentation $\mathcal{P}=\{X_1,\cdots,X_n\}$ if $x_{i,j}=\mu(\{\theta\in X_i:V(\theta)=v_j\})$ for all $i=1,\cdots,n$ and $j=1,2,\cdots,K$. It is obvious that for any $\{\mathbf{x}_1,\cdots,\mathbf{x}_n\}$,

there is a strategy σ that induces a market segmentation with canonical representation $\{\mathbf{x}_1,\cdots,\mathbf{x}_n\}$ if and only if $\sum_{i=1}^n x_{i.j} = \mu(\{\theta \in [0,1] : V(\theta) = v_j\}) \triangleq x_j^*$. Therefore, in the following construction, we focus on the canonical representation of market segmentation where $\sum_{i=1}^n \mathbf{x}_i = \mathbf{x}^*$.

The constructive approach necessarily requires non-uniform pricing, indicating that multiple optimal prices may exist in some markets (Lemma 1). Since we seek to implement the extreme case with $u = \overline{w} - \pi^*$, we enable multiple optimal prices in an extreme way: For every market within the segmentation, the producer is indifferent between charging any price inside the support, which coincides with the segmentation based on *extremal market* (Bergemann et al., 2015).[10]

For a set of valuations $S \subseteq \{v_1,\cdots,v_K\}$, the extremal market $\mathbf{x}^S(a)$ with total share being a, is defined as

$$x_i^S(a) = \begin{cases} 0 & v_i \notin S \\ a\min S\left(\frac{1}{v_i} - \frac{1}{\xi(v_i,S)}\right) & v_i \in S \end{cases}$$

where $\xi(v_i,S)$ denotes the smallest element in S higher than v_i, and $1/\xi(\max S,S) = 0$. We can verify that the producer is indifferent to charging any valuation that appears in an extremal market, resulting in a profit of $a\min S$. Hence, extremal markets are sensitive to entry and helpful for constructing strategy-proof segmentations.

With these good properties in mind, we generate markets iteratively. Recall that **supp** denotes the support set of a distribution. First, pack as many consumers as possible into an extremal market $\mathbf{x}^{\mathbf{supp}\{\mathbf{x}^*\}}(a_1)$ until running out of consumers with some valuation. The residual market is $\mathbf{x}^{(1)} = \mathbf{x}^* - \mathbf{x}^{\mathbf{supp}\{\mathbf{x}^*\}}(a_1)$. Second, pack as many consumers as possible from $\mathbf{x}^{(1)}$ into an extremal market $\mathbf{x}^{\mathbf{supp}\{\mathbf{x}^{(1)}\}}(a_2)$. The residual market is thus $\mathbf{x}^{(2)} = \mathbf{x}^{(1)} - \mathbf{x}^{\mathbf{supp}\{\mathbf{x}^{(1)}\}}(a_2)$. In each round k, we will obtain an extremal market $\mathbf{x}^{\mathbf{supp}\{\mathbf{x}^{(k-1)}\}}(a_k)$ and a residual market $\mathbf{x}^{(k)}$. This iteration is terminated when the remaining market becomes $\mathbf{0}$. Since $|\mathbf{supp}\{\mathbf{x}^{(i+1)}\}| < |\mathbf{supp}\{\mathbf{x}^{(i)}\}|$, there is at most K iterations, and the resulting segmentation is called *greedy segmentation* $\mathcal{P}^{\text{Greedy}}(\mathbf{x}^*) = \{\mathbf{x}_1^{\text{Greedy}},\cdots,\mathbf{x}_t^{\text{Greedy}},\mathbf{0},\cdots,\mathbf{0}\}$[11]. The greedy segmentation reaches $u = \overline{w} - \pi^*$. More importantly, the greedy segmentation is strategy-proof.

Remark 1. *The greedy segmentation $\mathcal{P}^{\text{Greedy}}(\mathbf{x}^*)$ is strategy-proof.*

This remark holds according to the following Lemma 3. Combining it with Lemma 1, we can derive a necessary and sufficient condition for a market to be strategy-proof under the producer strategy $\phi^{\min}$.

Lemma 3. *The strategy profile $(\phi^{\min},\sigma)$ induces a strategy-proof segmentation $\mathcal{P} = \{X_1,\cdots,X_n\}$ **if** the **indifference condition** holds.*

[10] Bergemann et al. (2015) construct two segmentations to implement $u = \overline{w} - \pi^*$: a segmentation based on direct market and a segmentation based on extremal market. In this paper, we formally prove that the latter is strategy-proof. However, the former one is not.

[11] Suppose the iteration lasts for t rounds, the remaining $n - t \geq 0$ markets are supplemented by $\mathbf{0}$.

Proof. See the Appendix. □

For sufficiency of Theorem 1, it remains to construct a family of strategy-proof segmentations that can achieve any intermediate $u \in (u^*, \overline{w} - \pi^*)$ based on greedy segmentation $\mathcal{P}^{\text{Greedy}}(\mathbf{x}^*)$. Given parameters $\alpha \in [0,1]$ and $k \in [1, t-1]$, we merge those markets that are generated later in greedy procedures to obtain a new segmentation, denoted by $\mathcal{P}^{\alpha,k}(\mathbf{x}^*)$:

$$\left\{ \underbrace{\mathbf{x}_1^{\text{Greedy}}, \cdots, \mathbf{x}_{k-1}^{\text{Greedy}}, \alpha \mathbf{x}_k^{\text{Greedy}}}_{\text{Extremal Markets}}, \underbrace{\sum_{j=k+1}^{t} \mathbf{x}_j^{\text{Greedy}} + (1-\alpha)\mathbf{x}_k^{\text{Greedy}}}_{\text{Last Market}}, \underbrace{\mathbf{0}, \cdots, \mathbf{0}}_{\text{Supplementary Markets}} \right\}.$$

Notably, the first k markets are extremal. Meanwhile, the minimum optimal price of the last market is v^*, since (i) v^* is optimal in every market generated by the greedy procedures, and (ii) v^* is the minimum optimal price in $\mathbf{x}_t^{\text{Greedy}}$ (Lemma 2).[12] Moreover, it is not hard to show that $\mathcal{P}^{\alpha,k}(\mathbf{x}^*)$ is strategy-proof by Lemma 3.

Remark 2. *The constructed segmentation $\mathcal{P}^{\alpha,k}(\mathbf{x}^*)$ is strategy-proof.*

Let Ψ denote the share of the last market, $\Psi = \left\| \sum_{j=k+1}^{t} \mathbf{x}_j^{\text{Greedy}} + (1-\alpha)\mathbf{x}_k^{\text{Greedy}} \right\|_1$. Given any $\Psi \in \left[\|\mathbf{x}_t^{\text{Greedy}}\|_1, 1 \right]$, there exists a unique $\mathcal{P}^{\alpha,k}(\mathbf{x}^*)$ such that the share of the last market is indeed Ψ. Meanwhile, the consumer surplus is continuous (and monotonic) in Ψ.[13] Since the consumer surplus is $\overline{w} - \pi^*$ when $\Psi = \|\mathbf{x}_t^{\text{Greedy}}\|_1$ and u^* when $\Psi = 1$, all consumer surplus within $[u^*, \overline{w} - \pi^*]$ can be achieved. This ends the proof of Theorem 1.

4 Deviation Costs

In this section, we assume that the deviation to another market incurs a cost of $c \geq 0$ for each consumer. Alternatively, c can capture the bounded rationality of consumers, in the sense that consumers cannot identify a profitable deviation unless that deviation improves by at least c. When $c = 0$, the benchmark analysis applies.

We use $\mathbf{SP}(c)$ to denote the possible set of surplus pairs (π, u) that can be achieved by strategy-proof segmentation given cost c. By our main result, $\mathbf{SP}(0) = \{(\pi^*, u) : u \in [u^*, \overline{w} - \pi^*]\}$. More-

[12] To see why the price of the last market is v^*, it suffices to show that (I) pricing any $v < v^*$ is strictly worse than v^* and (II) pricing any $v > v^*$ is not better than v^*. (I) holds since v^* is optimal in all markets and any $v < v^*$ is strictly worse than v^* in $\mathbf{x}_t^{\text{Greedy}}$ since $v^* = \phi^{\min}(\mathbf{x}_t^{\text{Greedy}})$. (II) holds immediately from (i).

[13] As Ψ increases by $\mathrm{d}\Psi$, those $\mathrm{d}\Psi$ consumers, once in the second last market $\alpha \mathbf{x}_k^{\text{Greedy}}$, face price v^* instead, while other consumers face the same price as before. Hence, the decline of consumer surplus is solely influenced by those changed consumers. Since changed consumers are distributed proportional to $\mathbf{x}_k^{\text{Greedy}}$, the loss of consumer surplus is bounded by $\mathrm{d}\Psi \left(v^* - \phi^{\min}(\mathbf{x}_k^{\text{Greedy}}) \right)$: Continuity holds.

over, the surplus triangle of Bergemann et al. (2015) can be interpreted as another extreme case: $\mathbf{SP}(\infty)=\{(\pi,u):\pi\geq\pi^*,u\geq 0,\pi+u\leq\overline{w}\}$.

We can directly get the following proposition.

Proposition 1. $\mathbf{SP}(c_1)\subseteq\mathbf{SP}(c_2)$ *when* $0\leq c_1\leq c_2$.

The proof of Proposition 1 is straightforward. Consider any strategy-proof segmentation with a deviation cost of c_1. By definition, no unilateral deviation exists where every consumer can improve their payoff by more than c_1. Consequently, for any higher deviation cost $c_2\geq c_1$, the same segmentation remains strategy-proof, as the incentive for deviation is further diminished.

Proposition 1 implies that, when the deviation cost is strictly positive, the set of potential welfare outcomes nests our result in the costless benchmark, but is nested by the surplus triangle of Bergemann et al. (2015). We use the following example to visualize this observation.

Example 1. *We consider an example such that there are three different reservation prices* $\{1,2,3\}$ *where* $V(\theta)=1$ *for* $\theta\in[0,\frac{1}{3}]$, $V(\theta)=2$ *for* $\theta\in(\frac{1}{3},\frac{2}{3}]$, *and* $V(\theta)=3$ *for* $\theta\in(\frac{2}{3},1]$. *We can calculate that* $\pi^*=\frac{4}{3},u^*=\frac{1}{3}$, *and* $\overline{w}=2$.

In this example, since the valuations have a minimum difference of 1, the deviation cost will influence the strategy-proof segmentation only if $c\geq 1$. Therefore, by Theorem 1, $\mathbf{SP}(c)=\{(\pi,u):\pi=\frac{4}{3},u\in[\frac{1}{3},\frac{2}{3}]\}$ for $c\in[0,1)$. When $1\leq c<2$, consumers with valuations 1 and 2 have no incentives to deviate. In this case, we only need to consider those consumers who have a valuation of 3 and are in a market with a price 3. We can calculate that $\mathbf{SP}(c)=\{(\pi,u):\frac{4}{3}\leq\pi\leq\frac{5}{3},0\leq u,u+\pi\leq 2]\}$ for $c\in[1,2)$. When $c\geq 2$, no consumers have incentives to deviate, so all segmentation must be strategy-proof, resulting in $\mathbf{SP}(c)=\{(\pi,u):\pi\geq\frac{4}{3},u\geq 0,\pi+u\leq 2\}$ for $c\geq 2$. We plot these sets in Figure 3 where AC represents the surplus pairs when the deviation cost is smaller than 1, the trapezoid $EFCD$ represents the surplus pairs when the deviation cost $c\in[1,2)$, and the whole triangle BCD represents the case where $c\geq 2$.

5 Comparison with Subgame Perfect Equilibrium

In this section, we study subgame-perfect equilibrium and show that strategy-proof market segmentation, as a solution concept, is a refinement of subgame-perfect equilibrium.

Definition 4. *A strategy profile* (ϕ,σ) *is a* ***subgame perfect equilibrium*** *if (i)* ϕ *is rational; (ii) for any* $\theta\in[0,1],k\in\{1,2,\cdots,n\}$, $u_\theta(\phi,\sigma(\theta),\sigma_{-\theta})\geq u_\theta(\phi,k,\sigma_{-\theta})$ *where* $\sigma_{-\theta}$ *is the strategy profile of all consumers except for consumer* θ.

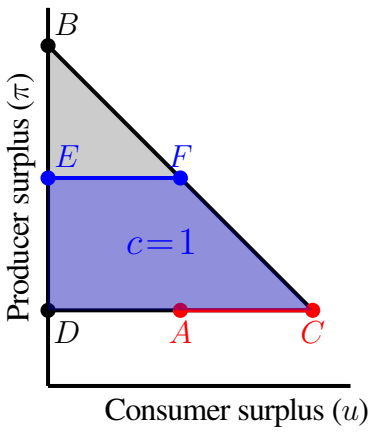

Figure 3: Welfare Consequences with Different Deviation Costs

The distinction between strategy-proof segmentation and subgame perfect equilibrium lies in their respective approaches to modeling individual deviations. In the context of strategy-proof segmentation, an individual's deviation constitutes a minimal deviation set, which consequently alters the canonical representation of market segmentation. In contrast, under the framework of subgame perfect equilibrium, an individual deviation does not modify the canonical representation of market segmentation. Yet, it should be noted that when multiple optimal prices exist within a market, the prevailing price may vary before and after a consumer's participation in one market.

In the following, we show that any strategy-proof segmentation $\mathcal{P}$ can be induced by a subgame-perfect equilibrium, but not vice versa. Thus, strategy-proof segmentation can be regarded as a refinement of the subgame perfect equilibrium.

Proposition 2. *For any strategy-proof market segmentation $\mathcal{P}$, there exists subgame-perfect equilibrium (ϕ,σ) that induces $\mathcal{P}$.*

Proof of Proposition 2. Given a strategy-proof segmentation $\mathcal{P}=(X_1,\cdots,X_n)$, we will construct a subgame-perfect equilibrium (σ,ϕ) such that the induced segmentation is $\mathcal{P}$.

Consumers' strategy σ satisfies that $\sigma(\theta)=k$ where $\theta\in X_k$. The producer's strategy ϕ is defined as follows. First, $\phi(X_i)\triangleq\phi^{\min}(X_i)$ for all i. Second, for a consumer $\theta\notin X_i$ with $V(\theta)=v_k$, we have $\phi(X_i\cup\{\theta\})\triangleq\lim_{\varepsilon\to 0^+}\phi^{\min}(\mathbf{x}_i+\varepsilon\mathbf{e}_k)$ where $\phi^{\min}(\mathbf{x})$ is the minimum optimal price of the market whose canonical representation is $\mathbf{x}$ and $\mathbf{e}_k$ is the K-dimensional vector that only the k-th component is non-zero and is equal to one. Third, for any other market X, $\phi(X)$ is any one of the optimal prices of market X.

Since $\lim_{\varepsilon\to 0^+}\phi^{\min}(\mathbf{x}_i+\varepsilon\mathbf{e}_k)$ is an optimal price of market X_i, ϕ is rational. Since ϕ takes an individual's deviation as a deviation set with infinitesimal size, once $\mathcal{P}$ is a strategy-proof segmentation, for any consumer θ, it cannot deviate to another market to obtain a higher payoff. □

However, a subgame perfect equilibrium might induce a market segmentation that is not strategy-proof. Using Example 1, we find a subgame perfect equilibrium under which the producer surplus is greater than π^*, as follows.

Consider the segmentation $\mathcal{P}=(X_1,X_2,\mathbf{0})$ with $X_1=[0,\frac{1}{3}]\cup(\frac{5}{6},1]$ and $X_2=(\frac{1}{3},\frac{5}{6}]$. The producer's strategy is $\phi(X_1)=1,\phi(X_2)=2$ and $\phi(X)$ is the maximum optimal price of X for markets $X\subseteq[0,1]$ other than X_1,X_2. Clearly, we can verify that both 1 and 3 are the optimal prices of market X_1, while 2 is the unique optimal price of market X_2. By Lemma 1, $\mathcal{P}$ is not strategy-proof. However, no consumers in market X_2 have an incentive to deviate to market X_1 since this change of X_1 will lead the price in market X_1 to be 3. Therefore, under the strategy ϕ, the segmentation survives under subgame perfect equilibrium. Now, the producer surplus is $\frac{3}{2}>\pi^*=\frac{4}{3}$.

6 Concluding Remarks

In this paper, we introduce a solution concept called strategy-proof market segmentation. This concept is proposed to capture the fact that each consumer remains non-negligible whenever moving between different market segments, even if the population of consumers is very large. Strategy-proof segmentation can be viewed as a refinement of subgame perfect equilibrium. Under a strategy-proof segmentation, producer surplus remains at the uniform monopoly level, while consumer surplus is at least the uniform monopoly level. Some strategy-proof segmentations can reach the efficient frontier.

However, there are some limitations regarding implementing our solution concept in practice. First, we assume that the producer is fully rational, a premise that may not hold in real-world scenarios. Specifically, we posit that the producer can accurately derive the exact value distribution of each market segment, and that her pricing algorithm can respond to even minor changes in this distribution by adjusting prices instantaneously. Second, we require that the prices set by the producer be discrete, meaning that any response from the producer must result in a price jump. If these assumptions are violated, the only feasible outcome collapses to the uniform monopoly outcome.

Nevertheless, with technological advancement, the aforementioned limitations are becoming less of a concern. Data-related services empower Internet platforms to aggregate consumer data, enabling real-time estimation of consumer value distribution within market segments. Furthermore, these platforms typically possess substantial computing power and advanced algorithms, which facilitate rational decision-making and allow for immediate responses to consumer actions.

Appendix: Proof of Lemma 3

Assume that the strategy profile $(\phi^{\min},\sigma)$ satisfies the indifference condition. We prove that the market segmentation must be strategy-proof. To establish this result, we employ a proof by contradiction.

Suppose there exists a market segmentation $\mathcal{P}=\{X_1,\cdots,X_n\}$ that satisfies the indifference condition, yet admits a profitable unilateral deviation. Without loss of generality, we assume $\phi(X_1)\leq\phi(X_2)\leq\cdots\leq\phi(X_n)$. Let r denote the minimum revenue difference between using the optimal price and using a non-optimal price in any market segment. For any $\varepsilon\in\left(0,\frac{r}{v_K-v_1}\right)$, there must exist a deviation set $D(\varepsilon)$ with $\mu(D(\varepsilon))\in(0,\varepsilon)$ and corresponding strategy profile σ' satisfying the conditions listed in Definition 1. We assume that $X_1',\cdots,X_n'$ is the market segmentation induced by σ'.

We first claim that $\phi^{\min}(X_i')$ must be an optimal price of market X_i. If not, pricing $\phi^{\min}(X_i)$ in market X_i' must yield higher profit than pricing $\phi^{\min}(X_i')$ by at least $r-\mu(D)(\phi^{\min}(X_i')-\phi^{\min}(X_i))>0$. This implies that $\phi^{\min}(X_i')$ is not an optimal price of market X_i', resulting in a contradiction. Since $\phi^{\min}(X_i')$ is an optimal price of X_i and $\phi^{\min}(X_i)$ is the minimum optimal price, we have $\phi^{\min}(X_i)\leq\phi^{\min}(X_i')$.

Let $L_i=D(\varepsilon)\cap X_i$ denote the set of consumers leaving X_i and $J_i=D(\varepsilon)\cap X_i'$ denote the set of consumers joining this market. Thus, $X_i'=(X_i\setminus L_i)\cup J_i$. Consider a market with the smallest index I such that $\mu(J_I)>0$. Since the price after deviation (i.e., $\phi^{\min}(X_I')$) will not decrease (i.e., $\phi^{\min}(X_I')\geq\phi^{\min}(X_I)$) and I is the smallest index with $\mu(J_I)>0$, we must have $\mu(L_I)=0$; otherwise, consumers in L_I must face a weakly higher price than $\phi^{\min}(X_I)$.

Let $v_k=\min\textbf{supp}\{J_I\}$ and we consider one consumer θ with valuation v_k coming from X_j, namely $\sigma(\theta)=j$. We must have $v_k>\phi^{\min}(X_I')\geq\phi^{\min}(X_I)$ to ensure that the consumer θ obtains a strictly positive payoff after the deviation. If $v_k\leq\phi^{\min}(X_j)$, then by the indifference condition, v_k is an optimal price of X_I; otherwise (i.e., $v_k>\phi^{\min}(X_j)$), $\phi^{\min}(X_j)$ is an optimal price of X_I and $\phi^{\min}(X_I')<\phi^{\min}(X_j)$ to ensure a profitable deviation. Thus, we know that $p=\min\{\phi^{\min}(X_j),v_k\}$ must be an optimal price of X_I and it is strictly larger than $\phi^{\min}(X_I')$. Finally, a contradiction arises,

$$\begin{aligned}0 &\geq p\int_{X_I'}\mathbb{I}\{V(\theta)\geq p\}d\theta-\phi^{\min}(X_I')\int_{X_I'}\mathbb{I}\{V(\theta)\geq\phi^{\min}(X_I')\}d\theta\\ &= (p-\phi^{\min}(X_I'))\mu(J_I)>0,\end{aligned}$$

where the former inequality is valid because $\phi^{\min}(X_I')$ is an optimal price of market X_I'; meanwhile, the latter inequality holds because $p>\phi^{\min}(X_I')$ and $\mu(J_I)>0$. As for the equation, let Π denote the revenue of pricing p or $\phi^{\min}(X_I')$ in market X_I. Then, the revenue of pricing p in market X_I' is $\Pi+p\mu(J_I)$ while the revenue of pricing $\phi^{\min}(X_I')$ in market X_I' is $\Pi+\phi^{\min}(X_I')\mu(J_I)$.

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